MUTTINENI PUJITHA

Full-Stack Developer · AI/ML Engineer · Generative AI · LLM Integration · Cloud Deployment

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PROFESSIONAL SUMMARY

Full-Stack Developer and AI/ML Engineer with 3+ years of end-to-end product development experience across ReactJS/Next.js frontends, Node.js/Python microservices, and production-grade Generative AI systems. Designed and deployed multi-agent LangGraph systems, agentic RAG pipelines, GAN-based generators, and LLM-integrated automation on AWS serverless infrastructure. Proven record of independent ownership from architecture through production deployment, with a security-first, performance-driven mindset. Google WE Scholar and Smart India Hackathon Finalist.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, SQL

Frontend: ReactJS, Next.js, React Native, Tailwind CSS, HTML5, CSS3, responsive/mobile-first UI

Backend: Node.js, Express.js, Python (FastAPI, Flask), REST API design, GraphQL, Microservices

AI/ML & GenAI: TensorFlow/Keras, PyTorch, scikit-learn, Hugging Face Transformers, GANs, ViLT, OpenCV, Rasa NLP

Agentic AI & LLMs: LangGraph, LangChain, CrewAI, RAG pipelines, FAISS & Pinecone, multi-agent supervisors, human-in-the-loop, Claude & OpenAI APIs, Prompt Engineering

Cloud & DevOps: AWS (Lambda, API Gateway, S3, EC2, DynamoDB), OCI, Docker, Kubernetes (GKE), GitHub Actions, Jenkins, Nginx, Linux, CI/CD

Databases: PostgreSQL, MongoDB, MySQL, DynamoDB, Firestore, Redis

Tools & Practices: Git, Postman, Jest/PyTest, Agile/Scrum, system design, schema & query optimisation

EXPERIENCE

Full-Stack + AI Developer | Lux Hospitals

- Building an AI-driven healthcare landing-page automation platform using Astro, Sanity CMS, React, and Generative AI to generate SEO-optimised campaign pages at scale.
- Engineered dynamic content generation and metadata automation pipelines, improving Core Web Vitals and page-load performance across production marketing surfaces.
- Integrated OpenAI and Claude APIs for AI-assisted content workflows, chatbot automation, and healthcare lead-generation systems.
- Owned AWS-based deployments, CI/CD pipelines, and full-stack platform architecture; partnered directly with founders on automation strategy and AI-first product direction.

Freelance Full-Stack Developer & AI/ML Engineer | Self-Employed

- Architected and shipped 4+ production-grade web applications with ReactJS/Next.js frontends and Node.js/Python backends — owning the full lifecycle from system design through AWS deployment independently.
- Built multi-agent systems with LangGraph and CrewAI, and agentic RAG pipelines using FAISS and Pinecone vector stores for enterprise-grade document reasoning and LLM-powered workflows.
- Deployed serverless microservices via AWS Lambda and API Gateway; built CI/CD pipelines with GitHub Actions and containerised services with Docker, enforcing Jest/PyTest coverage.
- Optimised PostgreSQL schemas and MongoDB queries for high-read workloads and applied Redis caching patterns, reducing API response times by ~40%.
- Accumulated 40+ hours of hands-on OCI Generative AI experience spanning model deployment, prompt engineering, and serverless inference patterns.

Full-Stack Developer Intern | CareAutomate

- Developed modular, mobile-responsive ReactJS interfaces for the mobile app, website, and marketing site serving US housing-market users — component-driven and production-quality.
- Built and secured Node.js/Express REST APIs backed by MongoDB with JWT authentication, request validation, and input sanitisation — security-first throughout.
- Collaborated with backend and ML teams on API schema design and data contracts, contributing to model-integrated feature development.

Frontend Developer Intern | One Convergence

- Delivered a full-stack Powerline Defect Detection system (ReactJS + Node.js + MongoDB), improving fault-detection accuracy by 35% and system scalability by 30%.
- Containerised and orchestrated ML microservices with Docker and Kubernetes on AWS; built Jenkins CI/CD pipelines that reduced Agile release cycles by 20%.

KEY AI/ML & FULL-STACK PROJECTS

InterviewForge AI Agent

- Architected a multi-agent system in LangGraph with specialised agents for DSA coaching, system-design feedback, behavioural simulation, and resume critique — tailored for FAANG-level prep.
- Built an agentic RAG pipeline (Pinecone + Claude API) retrieving role-specific questions from 500+ curated problems across Google, Meta, Amazon, Apple, and Netflix.
- Implemented adaptive difficulty logic that profiles user performance and dynamically adjusts complexity, cutting average preparation time by ~40% in self-evaluation benchmarks.
- Deployed FastAPI microservices on AWS Lambda with a React dashboard for progress tracking, mock sessions, and AI-generated feedback reports.

Enterprise Document Audit Agent

- Architected a 5-agent supervisor system in LangGraph to autonomously audit financial documents against SEBI/IFRS compliance rules, reducing manual review time by ~60%.
- Built an agentic RAG pipeline (FAISS + Claude API) for cross-document reasoning with human-in-the-loop approval gates.
- Deployed as a REST API on AWS Lambda + API Gateway with Docker; achieved <3s average latency and 92% clause-extraction accuracy.

GenAI Text-to-Image Generator

- Built an end-to-end text-to-image synthesis pipeline using GANs trained on OCI-hosted datasets; fine-tuned Vision-Language Transformers (ViLT) for accurate cross-modal alignment.
- Deployed trained models as REST APIs on OCI, applying model-optimisation techniques to reduce inference latency.

Flood-Prone Roads Identification System

- Designed and trained a CNN with TensorFlow and OpenCV for road-damage severity classification, deployed as a real-time full-stack alert system.
- Smart India Hackathon Finalist — recognised for an ML-powered flood-prediction solution with automated emergency alerts.

EDUCATION

B.Tech in Information Technology — CGPA: 9.37 / 10

ACHIEVEMENTS & CERTIFICATIONS

- **Smart India Hackathon Finalist** — built a full-stack ML solution predicting flood severity with real-time alerts to users and government officials.
- **Google WE Scholar '21** — selected by TalentSprint in collaboration with Google for outstanding potential in technology and innovation.
- **3rd Place, R&D Showcase 2024** — delivered an innovative AI-powered solution under tight deadlines.
- **Certifications:** Generative AI & Oracle Cloud Infrastructure (Oracle) · AWS Cloud Fundamentals · Database Programming with SQL (Oracle) · React Bootcamp (Scaler) · JavaScript (Educba) · CSS & HTML (LinkedIn Learning).